

LALIT KUMAR SAHOO

AI/ML Engineer

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PROFESSIONAL SUMMARY

Computer Science graduate focused on Machine Learning, Deep Learning, and Generative AI with hands-on experience using Python, PyTorch, Transformers, and machine learning frameworks. Experienced in building and experimenting with machine learning and AI applications, including model training, fine-tuning, and API-based deployment.

TECHNICAL SKILLS

Programming: Python, SQL

Data Science: NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Deep Learning: PyTorch, CNN, RNN, ANN

Generative AI / NLP: Transformers, Hugging Face, T5, Text Summarization, RAG, Embeddings, Vector Databases, Agentic AI

Backend / APIs: FastAPI, Django

Databases: MySQL, SQLite, ChromaDB

Tools: Git, GitHub, Jupyter Notebook, JupyterLab, VS Code, Postman

EDUCATION

B.Tech in Computer Science and Engineering Nalanda Institute of Technology, Odisha - 2024

Higher Secondary (12th) Bahanaga Higher Secondary School, Odisha – 59%

Secondary (10th) Sibaprasad High School, Odisha - 59%

PROJECTS

NovaGen Labs – Health Risk Prediction

Python | Pandas | Scikit-learn | Random Forest | GridSearchCV

- Built a classification model to predict health risk from physiological and lifestyle features.
- Applied data preprocessing, feature selection, train-test splitting, and model evaluation.
- Tuned Random Forest hyperparameters using 5-fold GridSearchCV with out-of-bag scoring.
- Achieved 88.80% test accuracy and 89.33% training accuracy.

TextSummarizer – Abstractive Text Summarization

Python | PyTorch | Hugging Face | Transformers | T5 | FastAPI

- Fine-tuned a pretrained T5 model for abstractive text summarization on a conversational dataset.
- Built a training pipeline using tokenization, sequence padding, gradient accumulation, and validation.
- Evaluated model performance during training and implemented beam-search based text generation.
- Developed a FastAPI backend to serve the trained model through a web application.

Loan Approval Prediction

Python | Pandas | Scikit-learn | KNN | Logistic Regression | Naive Bayes

- Built and evaluated Logistic Regression, KNN, and Gaussian Naive Bayes models for loan approval prediction after preprocessing applicant data.
- Compared models using accuracy, precision, recall, and F1-score to identify performance differences across classifiers.